

IDEATION PHASE

Brainstorm & Idea Prioritization

Date: 19 June 2026

Team ID: -----

Project Name: AI-Enhanced Intrusion Detection System

Maximum Marks: 4 Marks

Brainstorm & Idea Prioritization Template

Brainstorming provides a collaborative environment that encourages innovative thinking and problem-solving. Through structured idea generation and prioritization, the team identifies the most feasible and impactful solution for addressing cybersecurity challenges.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team Members

- Sourabh Bhimrao Patil
- Project Guide / Mentor
- Cybersecurity Domain Experts
- Machine Learning Contributors

Problem Statement

Traditional Intrusion Detection Systems (IDS) rely heavily on predefined signatures and rules, making them ineffective against emerging cyber threats and zero-day attacks. There is a need for an intelligent system capable of detecting, classifying, and responding to network intrusions in real time using Artificial Intelligence.

Diagram 1: Problem Statement Identification

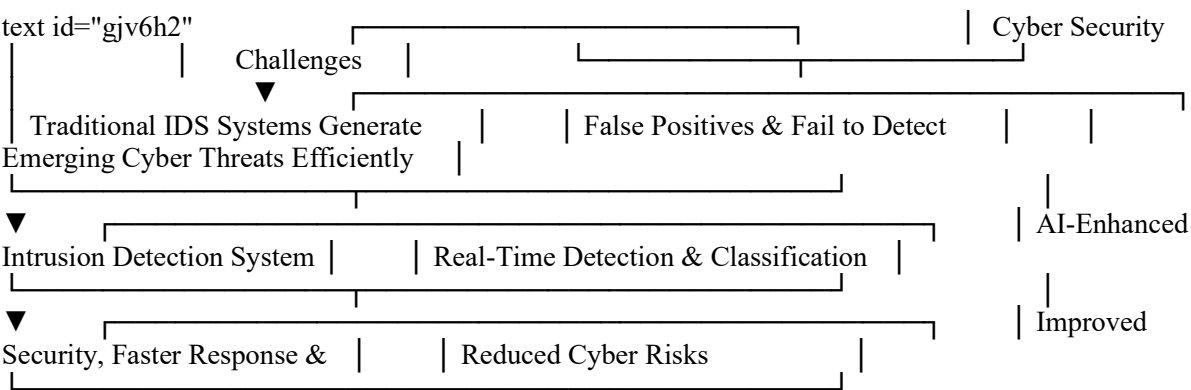


Figure 1: Problem Statement Selection

Step-2: Brainstorm, Idea Listing and Grouping

Brainstormed Ideas

Machine Learning Ideas

- Random Forest Intrusion Detection
- Decision Tree-Based Detection
- Support Vector Machine (SVM)
- Deep Learning-Based Detection

Security Monitoring Ideas

- Real-Time Threat Detection
- Automated Alert Management
- Attack Severity Classification

Data Processing Ideas

- NSL-KDD Dataset Analysis
- Feature Engineering
- Automated Data Preprocessing

Deployment Ideas

- REST API Integration
- Cloud-Based Deployment
- Security Analytics Dashboard

Idea Grouping

Group A – AI & Machine Learning

- Random Forest Classifier
- Support Vector Machine
- Deep Learning Models

Group B – Cybersecurity Monitoring

- Intrusion Detection
- Threat Classification
- Alert Management

Group C – User Interface

- Dashboard Monitoring
- Security Analytics
- Report Generation

Group D – Deployment & Integration

- REST API
- Cloud Infrastructure
- Real-Time Monitoring

Diagram 2: Brainstorming and Idea Grouping

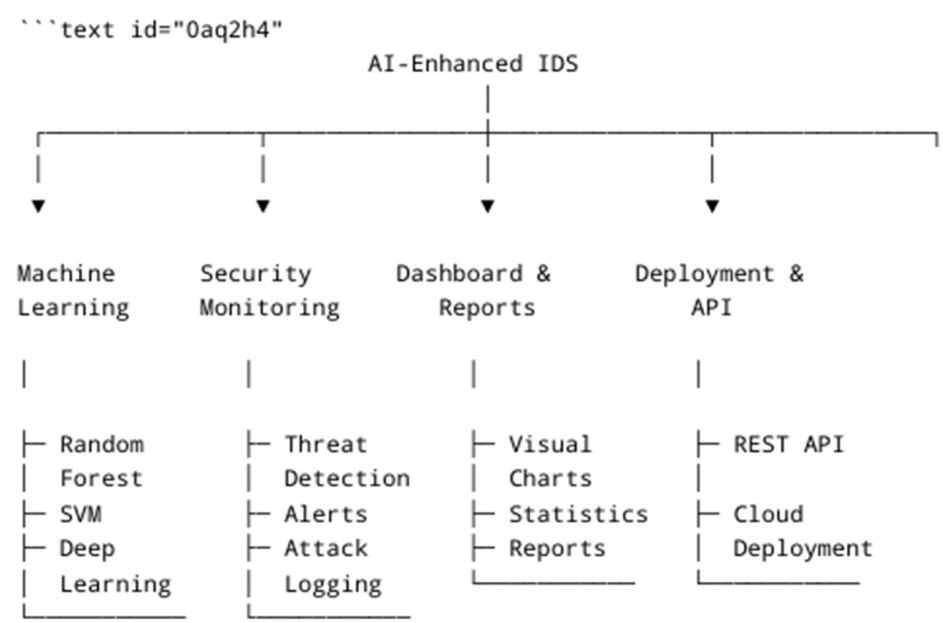


Figure 2: Brainstorming & Idea Grouping

Step-3: Idea Prioritization

The generated ideas were evaluated based on:

- 1. Importance
- 2. Feasibility
- 3. Development Complexity
- 4. Resource Requirements
- 5. Expected Impact

Prioritization Table

Idea	Importance	Feasibility	Priority Rank
Random Forest Intrusion Detection	High	High	1
Real-Time Threat Monitoring	High	High	2
Automated Alert System	High	High	3
Dashboard Analytics	High	High	4
REST API Integration	Medium	High	5
Cloud Deployment	Medium	Medium	6
Threat Intelligence Integration	Medium	Medium	7
Deep Learning-Based Detection	High	Low	8

Diagram 3: Idea Prioritization Matrix

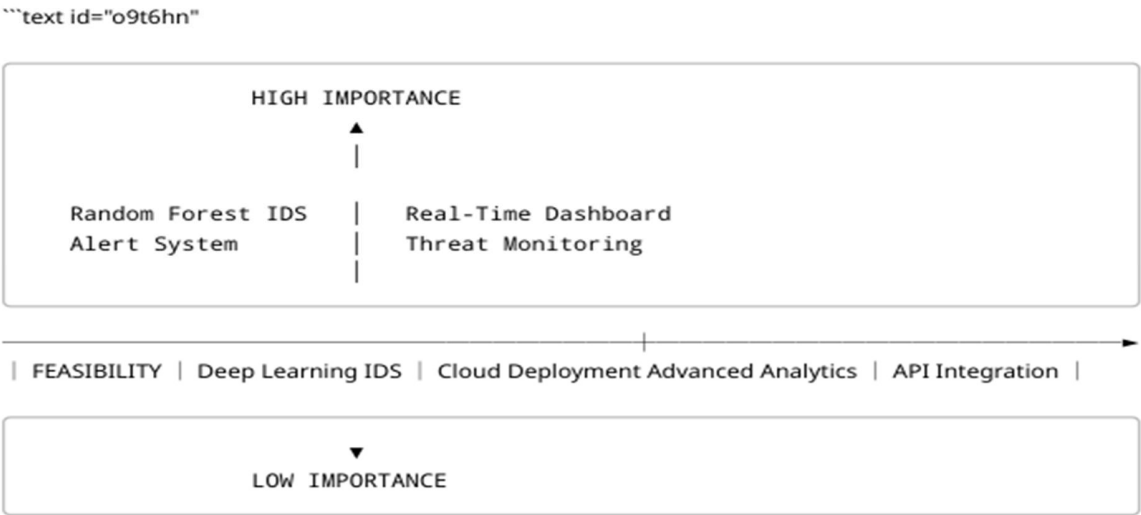


Figure 3: Idea Prioritization Matrix



Final Selected Idea

Selected Solution

AI-Enhanced Intrusion Detection System

The selected solution utilizes a Random Forest Machine Learning model trained on the NSL-KDD dataset to classify network traffic into:

- Normal Traffic
- DoS Attacks
- Probe Attacks
- R2L Attacks
- U2R Attacks

The system includes:

- Real-Time Intrusion Detection
- Automated Alert Logging
- Dashboard Monitoring
- REST API Support
- Security Analytics

Diagram 4: Final Solution Selection

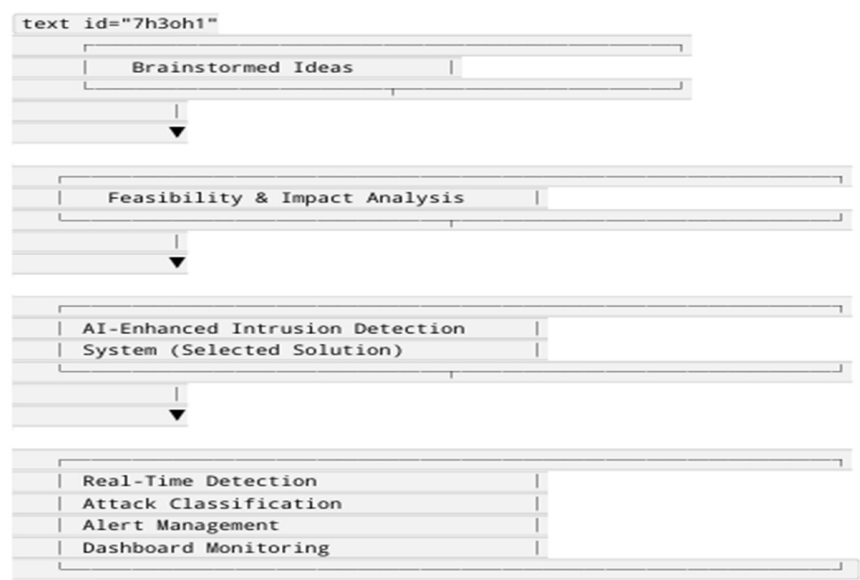


Figure 4: Final Selected Solution

Benefits of the Selected Solution

- High Detection Accuracy
 - Real-Time Threat Identification
 - Reduced False Positives
 - Automated Security Monitoring
 - Easy Integration with Existing Systems
 - Scalable Cloud Deployment Support
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Conclusion

Through brainstorming, idea grouping, and prioritization, the team identified the AI-Enhanced Intrusion Detection System as the most feasible and impactful solution. The selected approach combines Artificial Intelligence and Cybersecurity to provide real-time network threat detection, automated alert generation, and enhanced protection against modern cyber attacks.